JEE MAIN / PREVIOUS YEAR PAPER

JEE Main 2026 4th April Morning Shift

Question Paper with Answer Key

75	ENGLISH	DARK
QUESTIONS	LANGUAGE	FORMAT

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QR codes open the DekhoExams website.

Chemistry 25 questions

Q1. MCQ (Single Correct Answer)

Number of moles and number of molecules in 1.4187 L of SO_2 at STP respectively are

(A) $0.1266; 3.812 \times 10^{22}$

(B) $0.0633; 3.812 \times 10^{22}$

(C) $0.1266; 7.6238 \times 10^{22}$

(D) $0.0633; 7.6238 \times 10^{22}$

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Q2. MCQ (Single Correct Answer)

What is the ratio of wave number of first line (lowest energy line) of Balmer series of H atomic spectrum to first line of its Brackett series?

(A) 5 : 1

(B) 5 : 0.81

(C) 5 : 1.75

(D) 5 : 27

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Q3. MCQ (Single Correct Answer)

Which of the following is correct set of 4 quantum numbers of 19th electron in Chromium (Atomic number $= 24$) in accordance with Aufbau principle?

(A) $n = 3, l = 2, m = +2, s = +\frac{1}{2}$

(B) $n = 3, l = 2, m = -2, s = +\frac{1}{2}$

(C) $n = 4, l = 1, m = 0, s = +\frac{1}{2}$

(D) $n = 4, l = 0, m = 0, s = +\frac{1}{2}$

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Q4. MCQ (Single Correct Answer)

Given below are two statements :

Statement I : For an ideal gas, heat capacity at constant volume is always greater than the heat capacity at constant pressure.

Statement II : In a constant volume process, no work is produced and all the heat withdrawn goes into the chaotic motion and is reflected by a temperature increase of the ideal gas.

In the light of the above statements, choose the correct answer from the options given below

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true but Statement II is false

(D) Statement I is false but Statement II is true

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Q5. MCQ (Single Correct Answer)

At $\mathrm{T(K)}$, the equilibrium constant of

$\mathrm{A_2}(g) + \mathrm{B_2}(g) \rightleftharpoons \mathrm{C}(g)$ is 2.7×10^{-5}.

What is the equilibrium constant for

$\frac{1}{3}\,\mathrm{A_2(g)} + \frac{1}{3}\,\mathrm{B_2(g)} \rightleftharpoons \frac{1}{3}\mathrm{C(g)}$ at the same temperature?

(A) $\left(2.7 \times 10^{-5}\right)^3$

(B) 6×10^{-2}

(C) $\sqrt{2.7 \times 10^{-5}}$

(D) 3×10^{-2}

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Q6. MCQ (Single Correct Answer)

In order to oxidise a mixture of 1 mole each of $\mathrm{FeC_2O_4}, \mathrm{Fe_2(C_2O_4)_3}, \mathrm{FeSO_4}$ and $\mathrm{Fe_2(SO_4)_3}$ in acidic medium, the number of moles of $\mathrm{KMnO_4}$ required is

(A) 3

(B) 2

(C) 5

(D) 7

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Q7. MCQ (Single Correct Answer)

Consider the first order reaction $\mathrm{R} \rightarrow \mathrm{P}$.

The fraction of molecules decomposed in the given first order reaction can be expressed as

(A) $1 - e^{k_1 t}$

(B) $1 + e^{k_1 t}$

(C) $1 + e^{-k_1 t}$

(D) $1 - e^{-k_1 t}$

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Q8. MCQ (Single Correct Answer)

A monoatomic anion $(\mathrm{A^-})$ has 45 neutrons and 36 electrons. Atomic mass, group in the periodic table and physical state at room temperature of the element (A) respectively are

(A) 80, 17, liquid

(B) 81, 16, solid

(C) 80,16, gas

(D) 81, 15, gas

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Q9. MCQ (Single Correct Answer)

Given below are two statements :

Statement I : The covalency of oxygen is generally two but it can exceed upto four. The oxidation state of oxygen in $\mathrm{SO_2}$ is -2 and in $\mathrm{OF_2}$ it is +2 .

Statement II : The anomalous behaviour of oxygen when compared to the other elements of group 16 is due to its small size and high electronegativity.

In the light of the above statements, choose the correct answer from the options given below

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true but Statement II is false

(D) Statement I is false but Statement II is true

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Q10. MCQ (Single Correct Answer)

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The correct statements among the following are,

A. $\mathrm{Mo(VI)}$ and $\mathrm{W(VI)}$ are less stable than $\mathrm{Cr(VI)}$.

B. $\mathrm{Ce^{4+}}$ and $\mathrm{Tb^{4+}}$ are oxidant while $\mathrm{Eu^{2+}}$ and $\mathrm{Yb^{2+}}$ are reductant.

C. Cm and Am have seven unpaired electrons.

D. Actinoid contraction is greater from element to element than lanthanoid contraction.

Choose the correct answer from the options given below :

(A) A and B Only (B) C and D Only (C) B and D Only (D) A and C Only

Q11. MCQ (Single Correct Answer)

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Correct statements from the following are

A. Potassium dichromate is an oxidising agent and it oxidises $\mathrm{FeSO_4}$ to $\mathrm{Fe_2(SO_4)_3}$ in acidic medium.

B. Sodium dichromate can be used as primary standard in volumetric estimation.

C. $\mathrm{CrO_4^{2-}}$ and $\mathrm{Cr_2O_7^{2-}}$ are interconvertible in aqueous solution by varying the pH of the solution.

D. $\mathrm{Cr-O-Cr}$ bond angle in $\mathrm{Cr_2O_7^{2-}}$ is 126°.

Choose the correct answer from the options given below :

(A) A, B and C Only (B) A, C and D Only (C) A and C Only (D) B and D Only

Q12. MCQ (Single Correct Answer)

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Match the LIST-I with LIST-II

	List-I Complex ion		**List-II Calculated spin only magnetic moment (BM)**
A.	$[\mathrm{Cr(H_2O)_6}]^{2+}$	I.	3.87
B.	$[\mathrm{Co(H_2O)_6}]^{2+}$	II.	5.92
C.	$[\mathrm{Cu(H_2O)_6}]^{2+}$	III.	4.90
D.	$[\mathrm{Mn(H_2O)_6}]^{2+}$	IV.	1.73

Choose the correct answer from the options given below :

(A) A-I, B-III, C-IV, D-II (B) A-II, B-I, C-III, D-IV

(C) A-IV, B-II, C-I, D-III (D) A-III, B-I, C-IV, D-II

Q13. MCQ (Single Correct Answer)

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Increasing order of electron withdrawing power of following functional groups is :

a. -CN

b. -COOH

c. $-NO_2$

d. $-I$

(A) $c < b < d < a$

(B) $c < a < b < d$

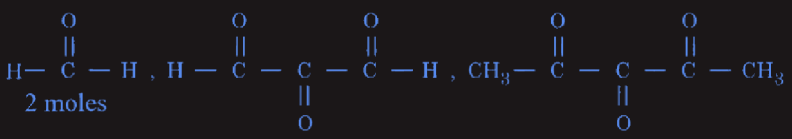
(C) $d < b < a < c$

(D) $a < b < c < d$

Q14. MCQ (Single Correct Answer)

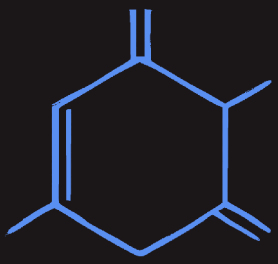
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An alkene (X) on ozonolysis followed by reduction gives following products.

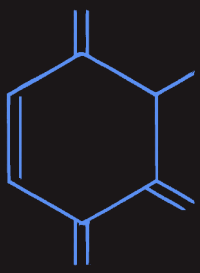


The alkene (X) is :

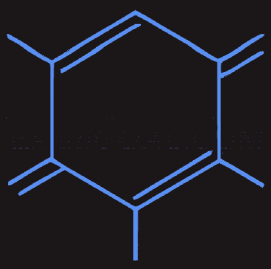
(A)



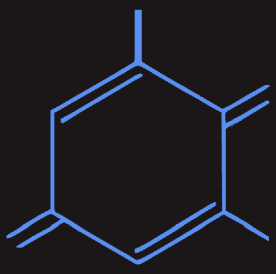
(B)



(C)



(D)



Q15. MCQ (Single Correct Answer)

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Match the LIST-I with LIST-II

List-I Name of reaction		List-II Reagent or catalyst used	
A.	Finkelstein reaction	I.	SbF_3
B.	Swarts reaction	II.	Na, dry ether
C.	Sandmeyer's reaction	III.	NaI
D.	Fittig reaction	IV.	Cu_2Cl_2

Choose the correct answer from the options given below :

(A) A-I, B-IV, C-III, D-II

(B) A-III, B-I, C-IV, D-II

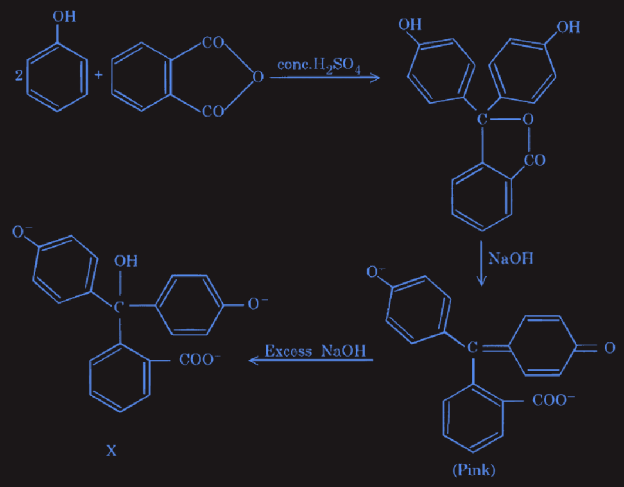
(C) A-IV, B-II, C-I, D-III

(D) A-I, B-III, C-II, D-IV

Q16. MCQ (Single Correct Answer)

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Identify the colour of compound 'X' in the sequence of the reaction.



(A) Violet

(B) Green

(C) Red

(D) Colourless

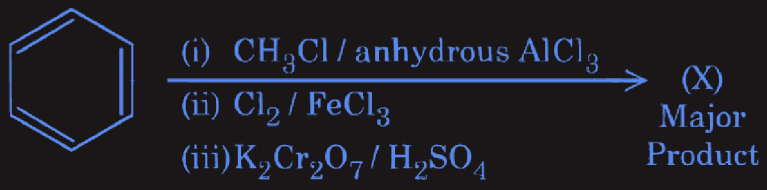
Q17. Numerical

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According to Lewis theory, the total number of σ bond-pairs and lone pair of electrons around the central atom of XeO_6^{4-} ion is ____ .

Q18. Numerical

Consider the following sequence of reactions to give the major product (X)



P g of the major product (X) formed is reacted with $NaHCO_3$ solution to liberate a gas which occupied $11.2 dm^3$ at STP.

$P =$ _____ g.

(Given molar mass in $gmol^{-1}$ $H : 1, C : 12, O : 16, Cl : 35.5$)

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Q19. Numerical

2.0 g of a bromo hydrocarbon (X) was subjected to Carius analysis, gave 3.36 g of AgBr . The percentage of carbon in the compound (X) is 26.7%. Total number of carbon atoms in the empirical formula for compound (X) is ____ .

(Given molar mass in $gmol^{-1}$ $H : 1, C : 12, Br : 80, Ag : 108$)

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Q20. Numerical

The pH of a solution obtained by mixing 5 mL of $0.1 M NH_4OH$ solution with 250 mL of $0.1 M NH_4Cl$ solution is ____ $\times 10^{-2}$.(Nearest integer) Given: $pK_b\ (NH_4OH) = 4.74$

$\log 2 = 0.30$
$\log 3 = 0.48$
$\log 5 = 0.70$

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Q21. Numerical

A non-volatile, non-electrolyte solid solute when dissolved in 40 g of a solvent, the vapour pressure of the solvent decreased from 760 mm Hg to 750 mm Hg . If the same solution boils at 320 K , then the number of moles of the solvent present in the solution is ____ . (Nearest integer)

[Given: boiling point of the pure solvent $= 319.5$ K,

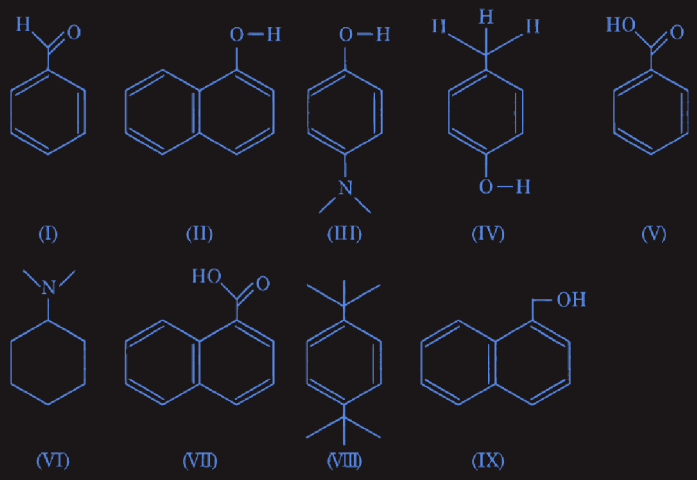
K_b of the solvent $= 0.3\ K\ kg\ mol^{-1}$]

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Q22. MCQ (Single Correct Answer)

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Amongst the following, the total number of compounds soluble in aqueous NaOH at room temperature is :

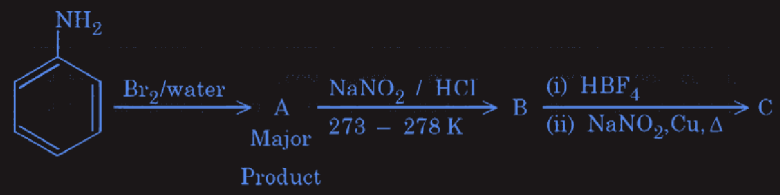


(A) 5 (B) 4 (C) 6 (D) 3

Q23. MCQ (Single Correct Answer)

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Product C of the following reaction sequence will be



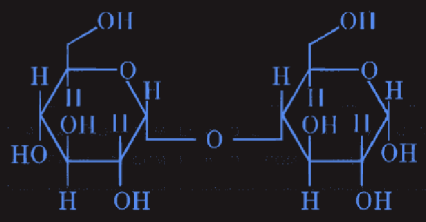
(A) 1-Bromo-4-nitrobenzene
(B) 1, 3, 5-Tribromo-2-nitrobenzene
(C) 4-Bromo-1-nitrobenzene
(D) 1, 3, 5-Tribromobenzene

Q24. MCQ (Single Correct Answer)

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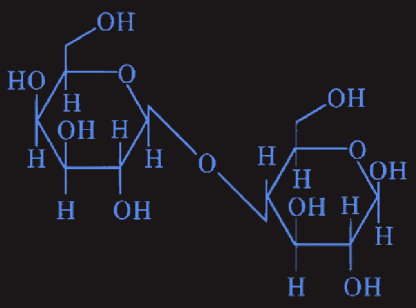
Given below are two statements :

Statement I : The structure of Maltose is given below:



Maltose is a non-reducing sugar.

Statement II : The structure of Lactose is given below :



Lactose is a reducing sugar.

In the light of the above statements, choose the **correct** answer from the options given below :

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true but Statement II is false

(D) Statement I is false but Statement II is true

Q25. MCQ (Single Correct Answer)

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Match the List-I with List-II

List-I **Name of amino acid**	**List-II** **One letter symbol/type**
A. Arginine	I. D/Non-essential
B. Aspartic acid	II. R/Essential
C. Lysine	III. E/Non-essential
D. Glutamic acid	IV. K/Essential

Choose the **correct** answer from the options given below :

(A) A-II, B-I, C-IV, D-III

(B) A-IV, B-III, C-II, D-I

(C) A-III, B-IV, C-I, D-II

(D) A-II, B-IV, C-I, D-III

Mathematics 25 questions

Q26. MCQ (Single Correct Answer)

Let [•] denote the greatest integer function. If the domain of the function

$f(x)=\cos^{-1}\left(\frac{4x+2[x]}{3}\right)$ is $[\alpha,\beta]$, then $12(\alpha+\beta)$ is equal to :

(A) 6 (B) 8 (C) 9 (D) 4

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Q27. MCQ (Single Correct Answer)

If the set of all solutions of $\left|x^2+x-9\right|=|x|+\left|x^2-9\right|$ is $[\alpha,\beta]\cup[\gamma,\infty)$, then $(\alpha^2+\beta^2+\gamma^2)$ is equal to:

(A) 9 (B) 18 (C) 36 (D) 72

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Q28. MCQ (Single Correct Answer)

Let z be a complex number such that $|z+2|=|z-2|$ and $\arg\left(\frac{z+3}{z-i}\right)=\frac{\pi}{4}$. Then $|z|^2$ is equal to:

(A) 9 (B) 4 (C) 5 (D) 1

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Q29. MCQ (Single Correct Answer)

The number of functions $f:\{1,2,3,4\}\to\{a,b,c\}$, which are not onto, is :

(A) 48 (B) 45 (C) 51 (D) 35

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Q30. MCQ (Single Correct Answer)

Let $\mathrm{S}=\left\{\mathrm{A}=\begin{bmatrix}a & b\\ c & d\end{bmatrix}: a,b,c,d\in\{0,1,2,3,4\}\text{ and }\mathrm{A}^2-4\,\mathrm{A}+3\mathrm{I}=0\right\}$ be a set of 2×2 matrices. Then the number of matrices in S , for which the sum of the diagonal elements is equal to 4 , is :

(A) 20 (B) 17 (C) 21 (D) 19

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Q31. MCQ (Single Correct Answer)

Let $A=\begin{bmatrix}1 & 1 & 2\\ -2 & 0 & 1\\ 1 & 3 & 5\end{bmatrix}$. Then the sum of all elements of the matrix $\operatorname{adj}\left(\operatorname{adj}\left(2(\operatorname{adj}\ \mathrm{A})^{-1}\right)\right)$ is equal to:

(A) 3 (B) 4 (C) -4 (D) -3

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Q32. MCQ (Single Correct Answer)

The first term of an A.P. of 30 non-negative terms is $\frac{10}{3}$. If the sum of this A.P. is the cube of its last term, then its common difference is:

(A) $\frac{5}{87}$ (B) $\frac{25}{83}$ (C) $\frac{15}{29}$ (D) $\frac{5}{29}$

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Q33. MCQ (Single Correct Answer)

The number of ways, of forming a queue of 4 boys and 3 girls such that all the girls are not together, is:

(A) 5040 (B) 3050 (C) 3410 (D) 4320

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Q34. MCQ (Single Correct Answer)

Let the smallest value of $k \in \mathbb{N}$, for which the coefficient of x^3 in $(1+x)^3+(1+x)^4+(1+x)^5+\ldots+(1+x)^{99}+(1+kx)^{100}, x \neq 0$, is $\left(43n+\frac{101}{4}\right)\left({}^{100}C_3\right)$ for some $n \in \mathbb{N}$, be p. Then the value of $p+n$ is :

(A) 10 (B) 11 (C) 12 (D) 13

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Q35. MCQ (Single Correct Answer)

Suppose that the mean and median of the non-negative numbers $21, 8, 17, a, 51, 103, b, 13, 67, (a > b)$, are 40 and 21 , respectively. If the mean deviation about the median is 26 , then $2a$ is equal to :

(A) 109 (B) 117 (C) 161 (D) 131

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Q36. MCQ (Single Correct Answer)

Let the line $L_1 : x+3=0$ intersect the lines $L_2 : x-y=0$ and $L_3 : 3x+y=0$ at the points A and B , respectively. Let the bisector of the obtuse angle between the lines L_2 and L_3 intersect the line L_1 at the point C. Then $BC^2 : AC^2$ is equal to:

(A) 5:1 (B) 1:5 (C) 2:3 (D) 3:2

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Q37. MCQ (Single Correct Answer)

Let the vertex A of a triangle ABC be $(1, 2)$, and the mid-point of the side AB be $(5, -1)$. If the centroid of this triangle is $(3, 4)$ and its circumcenter is (α, β), then $21(\alpha+\beta)$ is equal to :

(A) 309 (B) 403 (C) 497 (D) 524

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Q38. MCQ (Single Correct Answer)

Suppose that two chords, drawn from the point $(1, 2)$ on the circle $x^2+y^2+x-3y=0$ are bisected by the y-axis. If the other ends of these chords are R and S , and the mid point of the line segment RS is (α, β), then $6(\alpha+\beta)$ is equal to :

(A) 1 (B) 3 (C) 4 (D) 6

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Q39. MCQ (Single Correct Answer)

A line with direction ratios $1, -1, 2$ intersects the lines $\frac{x}{2} = \frac{y}{3} = \frac{z+1}{3}$ and $\frac{x+1}{-1} = \frac{y-2}{1} = \frac{z}{4}$ at the points P and Q , respectively. If the length of the line segment PQ is α, then $225\alpha^2$ is equal to:

(A) 1024 (B) 1014 (C) 1104 (D) 1204

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Q40. MCQ (Single Correct Answer)

The square of the distance of the point $(-2, -8, 6)$ from the line $\frac{x-1}{1} = \frac{y-1}{2} = \frac{z}{-1}$ along the line $\frac{x+5}{1} = \frac{y+5}{-1} = \frac{z}{2}$ is equal to:

(A) 3 (B) 6 (C) 8 (D) 12

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Q41. MCQ (Single Correct Answer)

If $y = \tan^{-1}\left(\frac{3\cos x - 4\sin x}{4\cos x + 3\sin x}\right) + 2\tan^{-1}\left(\frac{x}{1+\sqrt{1-x^2}}\right)$, then $\frac{dy}{dx}$ at $x = \frac{\sqrt{3}}{2}$ is equal to :

(A) 3 (B) -1 (C) 1 (D) 2

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Q42. MCQ (Single Correct Answer)

Let f be a real polynomial of degree n such that $f(x) = f'(x)f''(x)$, for all $x \in \mathbb{R}$. If $f(0) = 0$, then $36\left(f'(2) + f''(2) + \int_0^2 f(x)dx\right)$ is equal to:

(A) 42 (B) 46 (C) 56 (D) 66

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Q43. MCQ (Single Correct Answer)

The area of the region $\{(x, y) : y \le \pi - |x|, y \le |x\sin x|, y \ge 0\}$ is:

(A) $1 + \frac{\pi^2}{8}$ (B) $2 + \frac{\pi^2}{4}$ (C) $\frac{\pi^2}{8} - 1$ (D) $4 + \frac{\pi^2}{2}$

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Q44. MCQ (Single Correct Answer)

Let $\int_{-2}^{2}(|\sin x| + [x\sin x])dx = 2(3 - \cos 2) + \beta$, where [·] is the greatest integer function. Then $\beta\sin\left(\frac{\beta}{2}\right)$ equals:

(A) 1 (B) 2 (C) 4 (D) 8

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Q45. MCQ (Single Correct Answer)

Let $y = y(x)$ be the solution of the differential equation $\frac{dy}{dx} = \left(1 + x + x^2\right)\left(1 - y + y^2\right), y(0) = \frac{1}{2}$. Then $(2y(1) - 1)$ is equal to

(A) $\sqrt{3}\tan\left(\frac{11\sqrt{3}}{6}\right)$ (B) $\frac{\sqrt{3}}{2}\tan\left(\frac{11\sqrt{3}}{12}\right)$

(C) $\sqrt{3}\tan\left(\frac{11\sqrt{3}}{12}\right)$ (D) $\frac{\sqrt{3}}{2}\tan\left(\frac{11\sqrt{3}}{6}\right)$

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Q46. Numerical

A coin is tossed 8 times. If the probability that exactly 4 heads appear in the first six tosses and exactly 3 heads appear in the last five tosses is p, then $96p$ is equal to ____ .

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Q47. Numerical

Consider the parabola $\mathrm{P}: y^2 = 4kx$ and the ellipse $\mathrm{E}: \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. Let the line segment joining the points of intersection of P and E , be their latus rectums. If the eccentricity of E is e, then $e^2 + 2\sqrt{2}$ is equal to ____ .

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Q48. Numerical

If $\mathrm{A} = \frac{\sin 3^\circ}{\cos 9^\circ} + \frac{\sin 9^\circ}{\cos 27^\circ} + \frac{\sin 27^\circ}{\cos 81^\circ}$ and $\mathrm{B} = \tan 81^\circ - \tan 3^\circ$, then $\frac{\mathrm{B}}{\mathrm{A}}$ is equal to ____ .

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Q49. Numerical

Let $\overrightarrow{a_k} = (\tan\theta_k)\hat{i} + \hat{j}$ and $\overrightarrow{b_k} = \hat{i} - (\cot\theta_k)\hat{j}$, where $\theta_k = \frac{2^{k-1}\pi}{2^n+1}$, for some $n \in \mathbb{N}, n > 5$. Then the value of $\frac{\sum_{k=1}^{n}\left|\overrightarrow{a_k}\right|^2}{\sum_{k=1}^{n}\left|\overrightarrow{b_k}\right|^2}$ is

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Q50. Numerical

The number of points, at which the function
$f(x) = \max\left\{6x, 2+3x^2\right\} + |x-1|\cos\left|x^2 - \frac{1}{4}\right|, x \in (-\pi, \pi)$, is not differentiable, is ____ .

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Physics 24 questions

Q51. MCQ (Single Correct Answer)

In a screw gauge when the circular scale is given five complete rotations it moves linearly by 2.5 mm . If the circular scale has 100 divisions, the least count of screw gauge is ____ mm.

(A) 1×10^{-2} (B) 1×10^{-3} (C) 5×10^{-2} (D) 5×10^{-3}

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Q52. MCQ (Single Correct Answer)

The increase in the pressure required to decrease the volume (ΔV) of water is $6.3 \times 10^7\ \mathrm{N/m^2}$. The percentage decrease in the volume is ____ .

(Bulk modulus of water $= 2.1 \times 10^9\ \mathrm{N/m^2}$.)

(A) 2 % (B) 3 % (C) 6 % (D) 4 %

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Q53. MCQ (Single Correct Answer)

The time taken by a block of mass m to slide down from the highest point to the lowest point on a rough inclined plane is 50% more compared to the time taken by the same block on identical inclined smooth plane. Both inclined planes are at 45° with the horizontal. The coefficient of kinetic friction between the rough inclined surface and block is ____

(A) 3/4 (B) 2/3 (C) 5/9 (D) 4/9

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Q54. MCQ (Single Correct Answer)

Two nuclei of mass number 3 combine with another nucleus of mass number 4 to yield a nucleus of mass number 10. If the binding energy per nucleon for the mass numbers 3,4 and 10 are $5.6\mathrm{MeV}, 7.4\mathrm{MeV}$ and 6.1 MeV , respectively, then in the process, $\Delta \mathrm{Mc}^2 =$ ____ MeV .

(A) 6.9 (B) 7.9 (C) 2.2 (D) 4.3

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Q55. MCQ (Single Correct Answer)

A solid sphere of mass M and radius R is divided into two unequal parts. The smaller part having mass $M/8$ is converted into a sphere of radius r and the larger part is converted into a circular disc of thickness t and radius $2R$. If I_1 is moment of inertia of a sphere having radius r about an axis through its centre and I_2 is the moment of inertia of a disc about its diameter, the ratio of their moment of inertia $I_2/I_1 =$ ____

(A) 35 (B) 70 (C) 140 (D) 210

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Q56. MCQ (Single Correct Answer)

The two projectiles are projected with the same initial velocities at the 15° and 30° with respect to the horizontal. The ratio of their ranges is $1 : x$. The value of x is

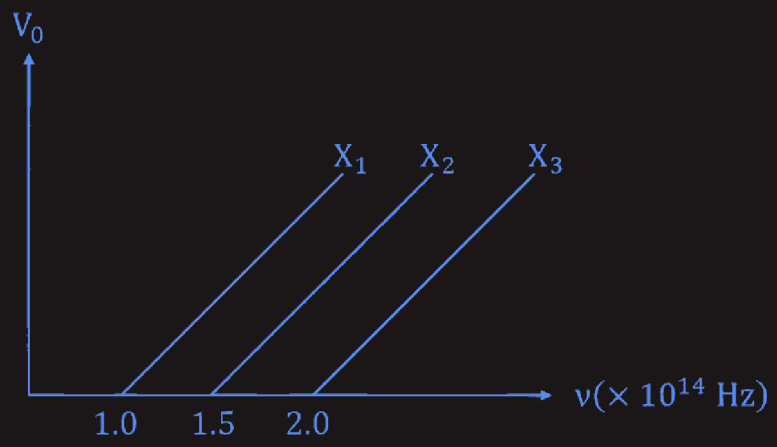
(A) $\sqrt{2}$ (B) $\sqrt{3}$ (C) $2\sqrt{3}$ (D) $\frac{1}{\sqrt{2}}$

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Q57. MCQ (Single Correct Answer)

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The graph shows variation of stopping potential V_o with the frequency v of the incident radiation for three photosensitive metals X_1, X_2 and X_3. Which metal will give out electrons with greater kinetic energy, for the same wavelength of incident radiation?



(A) X_1

(B) X_2

(C) X_3

(D) All the metals will give out photo electrons with same kinetic energies.

Q58. MCQ (Single Correct Answer)

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A slit of width a is illuminated by light of wavelength λ. The linear separation between 1st and 3rd minima in the diffraction pattern produced on a screen placed at a distance D from the slit system is ____ .

(A) $\frac{D\lambda}{a}$ (B) $1.5\frac{D\lambda}{a}$ (C) $2\frac{D\lambda}{a}$ (D) $3\frac{D\lambda}{a}$

Q59. MCQ (Single Correct Answer)

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A string A of length 0.314 m and Young's modulus $2 \times 10^{10}\ \mathrm{N/m^2}$ is connected to another string B of length and Young's modulus both twice of those of A. This series combination of strings is then suspended from a rigid support and its free end is fixed to a load of mass 0.8 kg . The net change in length of the combination is ____ mm.

(radius of both the strings is 0.2 mm and acceleration due to gravity $= 10\ \mathrm{m/s^2}$) (Mass of both strings is to be neglected as compared to the mass of load)

(A) 3 (B) 2 (C) 1.9 (D) 1

Q60. MCQ (Single Correct Answer)

One gas of n_1 mole of molecules at temperature T_1, volume V_1, and pressure P_1, and another gas of n_2 mole of molecules at temperature T_2, volume V_2, and pressure P_2, are mixed resulting in pressure P and volume V of the mixture. The temperature of the mixture is ____ .

(A) $\frac{(T_1+T_2)}{2}$

(B) $\frac{T_1T_2PV}{T_2P_1V_1+T_1P_2V_2}$

(C) $\frac{(T_2P_1V_1+T_1P_2V_2)}{T_1T_2PV}$

(D) $\frac{[T_1-T_2]}{2}$

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Q61. MCQ (Single Correct Answer)

An ideal gas undergoes a process maintaining relation between pressure (P) and $\text{volume}(V)$ as $P = P_o\left(1+\left(\frac{V_o}{V}\right)^2\right)^{-1}$, where P_o and V_o are constants. If two samples A and B (two moles each) with initial volumes V_o and $3V_o$ respectively undergo above mentioned process and attain same pressure, then the difference at the temperatures of these samples, $T_B - T_A$ is ____ .

($R =$ gas constant)

(A) $\frac{9P_oV_o}{8R}$

(B) $\frac{11P_oV_o}{10R}$

(C) $\frac{7P_oV_o}{6R}$

(D) $\frac{13P_oV_o}{11R}$

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Q62. MCQ (Single Correct Answer)

A voltmeter with internal resistance of $x\Omega$ can be used to measure upto 20 V . In order to increase its measuring range to 30 V , the required modification is to ____ .

(A) connect resistor of $\frac{x}{2}\Omega$, in series with voltmeter.

(B) connect resistor of $\frac{x}{2}\Omega$, in parallel to voltmeter.

(C) connect a resistor of $x\Omega$ in series with voltmeter.

(D) connect resistor of $2x\Omega$ in parallel to voltmeter.

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Q63. MCQ (Single Correct Answer)

Two 4 bits binary numbers, $A = 1101$ and $B = 1010$ are given in the inputs of a logic circuit shown in figure below. The output (Y) will be :



(A) $Y = 1101$

(B) $Y = 0010$

(C) $Y = 0111$

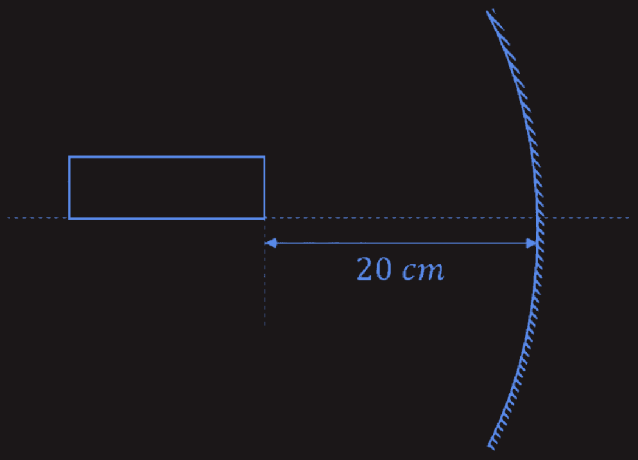
(D) $Y = 1000$

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Q64. MCQ (Single Correct Answer)

A rod of length 10 cm lies along the principle axis of a concave mirror of focal length 10 cm as shown in figure. The length of the image is ____ cm.

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(A) 2.5 (B) 5 (C) 7.5 (D) 7

Q65. MCQ (Single Correct Answer)

A parallel plate air capacitor is connected to a battery. The plates are pulled apart at uniform speed v. If x is the separation between the plates at any instant, then the time rate of change of electrostatic energy of the capacitor is proportional to x^{α}, where α is ____ .

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(A) -2 (B) 1 (C) -1 (D) 2

Q66. MCQ (Single Correct Answer)

An insulated wire is wound so that it forms a flat coil with $N = 200$ turns. The radius of the innermost turn is $r_1 = 3\ \text{cm}$, and of the outermost turn $r_2 = 6\ \text{cm}$. If 20 mA current flows in it then the magnetic moment will be $\alpha \times 10^{-2}\ \text{A.m}^2$. The value of α is ____ .

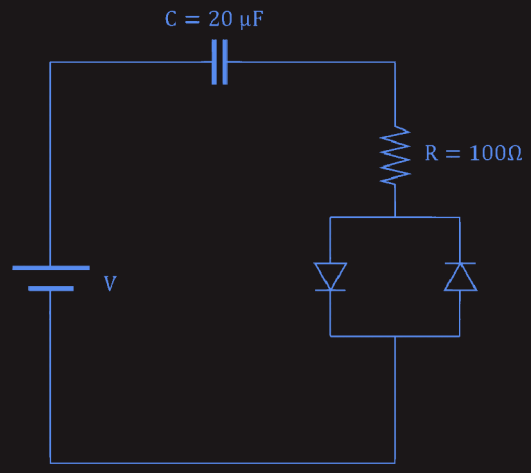
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(A) 4.4 (B) 2.64 (C) 3.25 (D) 1.2

Q67. MCQ (Single Correct Answer)

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Consider a circuit consisting of a capacitor $(20\mu\,\mathrm{F})$, resistor (100Ω) and two identical diodes as shown in figure. The resistance of diode under forward biasing condition is 10Ω. The time constant of the circuit is $\alpha \times 10^{-3}$ s. The value of α is ____

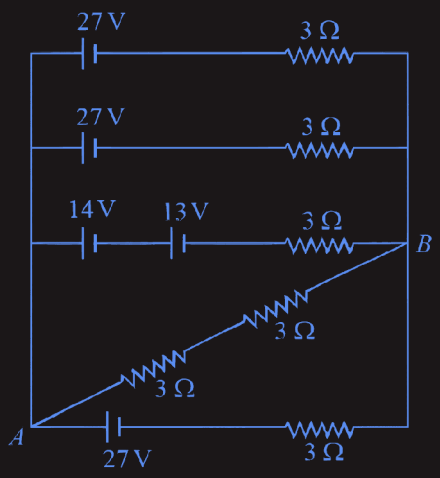


(A) 2.2 (B) 2.0 (C) 2.1 (D) 2.4

Q68. MCQ (Single Correct Answer)

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The voltage and the current between A and B points shown in the circuit are ____ .

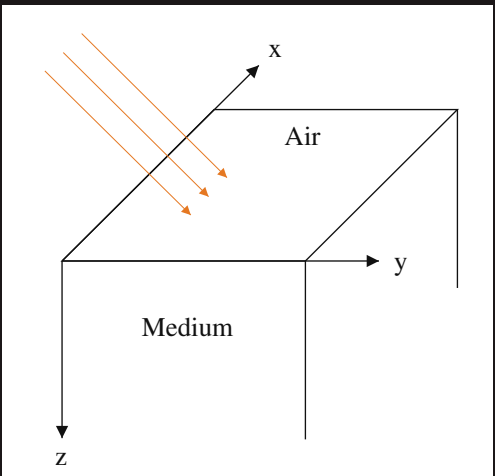


(A) $24\,\mathrm{V}, 12\,\mathrm{A}$ (B) $24\,\mathrm{V}, 4\,\mathrm{A}$ (C) $18\,\mathrm{V}, 12\,\mathrm{A}$ (D) $27\,\mathrm{V}, 4\,\mathrm{A}$

Q69. MCQ (Single Correct Answer)

An unpolarized light is incident on the plane interface of air-dielectric medium shown in figure. If the incident angle is equal to Brewster angle, identify the expression representing reflected wave.

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(A) $\left(E_x\hat{i}+E_y\hat{j}\right)\sin(kx-kz-\omega t)$

(B) $\left(E_x\hat{i}+E_z\hat{k}\right)\sin(kx+ky-\omega t)$

(C) $\left(E_x\hat{j}+E_y\hat{k}\right)\sin(ky+kz-\omega t)$

(D) $\left(E_x\hat{i}+E_y\hat{j}+E_z\hat{k}\right)\sin(kx+ky-kz-\omega t)$

Q70. Numerical

A 1 kg block subjected to two simultaneous forces $(2\hat{\mathrm{i}}+3\hat{\mathrm{j}}+4\hat{\mathrm{k}})\mathrm{N}$ and $(3\hat{\mathrm{i}}-\hat{\mathrm{j}}-2\hat{\mathrm{k}})\mathrm{N}$ is moved a distance of 25 m along $(3\hat{\mathrm{i}}-4\hat{\mathrm{j}})$ direction. The work done in this process is ____ J.

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Q71. Numerical

The surface tension of a soap solution is $3.5\times10^{-2}\,\mathrm{N/m}$. The work required to increase the radius of a soap bubble from 1 cm to 2 cm is $\alpha\times10^{-6}$ J. The value of α is ____ .

$(\pi=22/7)$

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Q72. Numerical

The velocity of a particle executing simple harmonic motion along x-axis is described as $v^2=50-x^2$, where x represents displacement. If the time period of motion is $\frac{x}{7}$ s, the value of x is ____ .

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Q73. Numerical

A body of mass 2 kg begins to move under the influence of time dependent force $\vec{\mathrm{F}}=\left(2\mathrm{t}\hat{\mathrm{i}}+6\mathrm{t}^2\hat{\mathrm{j}}\right)\mathrm{N}$, where $\hat{\mathrm{i}}$ and $\hat{\mathrm{j}}$ are unit vectors along x and y -axis respectively. The power produced by the force at $t=2\,\mathrm{s}$ is ____ W.

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Q74. Numerical

An inductor of 10 mH , capacitor of $0.1\mu\,\mathrm{F}$ and a resistor of 100Ω are connected in series across an $a.c$ power supply $220\ \mathrm{V}, 70\ \mathrm{Hz}$. The power factor of the given circuit is 0.5 . The difference in the inductive reactance and capacitance reactance is $\sqrt{3}\alpha\Omega$. The value of α is ____ .

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Out of syllabus 1 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q75. MCQ (Single Correct Answer) · **Physics**

A telescope with objective diameter R is used to observe a distant star emitting light of wavelength 500 nm , at a resolution of 5×10^{-7} radian. The value of R is

____ cm .

(A) 61 (B) 122 (C) 244 (D) 305

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Answer Key

Chemistry

1. **B**	2. **B**	3. **D**	4. **D**	5. **D**
6. **B**	7. **D**	8. **A**	9. **A**	10. **C**
11. **B**	12. **D**	13. **C**	14. **D**	15. **B**
16. **D**	17. **6**	18. **78**	19. **5**	20. **756**
21. **5**	22. **A**	23. **B**	24. **D**	25. **A**

Mathematics

26. **A**	27. **B**	28. **A**	29. **B**	30. **D**
31. **D**	32. **A**	33. **D**	34. **B**	35. **D**
36. **A**	37. **C**	38. **B**	39. **B**	40. **B**
41. **C**	42. **C**	43. **B**	44. **B**	45. **C**
46. **9**	47. **3**	48. **2**	49. **3**	50. **3**

Physics

51. **D**	52. **B**	53. **C**	54. **C**	55. **B**
56. **B**	57. **A**	58. **C**	59. **B**	60. **B**
61. **B**	62. **A**	63. **A**	64. **B**	65. **A**
66. **B**	67. **A**	68. **B**	69. **A**	70. **35**
71. **264**	72. **44**	73. **200**	74. **100**	

Out of syllabus

75. **B**	

For numerical answers given as a range, any value within the range is correct.

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01 Attempt the paper

Work through the questions before checking the answers.

02 Check the answer key

Compare your responses with the key included in this PDF.

03 Review your mistakes

Revisit the concepts behind the questions you missed.

04 Practise again

Return to difficult questions and track your improvement.

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